Aviat NETWORKS

WTM 4000

MISSION CRITICAL ALL-OUTDOOR, IP/SDN RADIO

DATASHEET [ANSI – U.S. GOVERNMENT]

NTIA TAA COMPLIANT FIPS 140-3™ DISA DoDIN APL

The WTM 4000 platform delivers ultra-high capacity in single or dual-transceiver architecture, optimized for all-outdoor applications. With up to 2.5 Gbit/s of throughput in a single unit, WTM 4000 supports all microwave bands with cutting edge networking features to set the benchmark for next generation backhaul and transport requirements.

Comprising WTM 4100, WTM 4200 and WTM 4500, the WTM 4000 series includes the very latest capacity innovations, including 4096 QAM, 80 MHz channels, Adaptive Dual Carrier and Multi Layer Header Compression, with single and dual transceiver configurations, all in a compact all outdoor device.

With a highly secure and self-contained architecture, integrated CE/MPLS-capable switch and hardware redundancy, WTM 4000 supports best-in-class system gain performance for longer reach, smaller antennas, improved link availability and lower TCO.

WTM 4000 exceeds industry standards for security, with Authentication Authorisation and Accounting (AAA), payload encryption and secure management. FIPS 140-3 validation means WTM 4000 meets Government-mandated security policies for data transport over microwave radio networks.

Maximum capacities up to 2.5 Gbit/s per link in microwave bands from 5 to 42 GHz, with full hardware redundancy for robust and reliable backhaul.

WTM 4000 is the only AODR platform validated to FIPS 140-3, providing ultimate security assurance while meeting Government requirements.

WTM 4000 provides a powerful microwave networking solution allowing "transparent" connections, or comprehensive Carrier Ethernet (CE), or IP/MPLS operation.

PROUDLY MANUFACTURED IN THE USA

General Specifications

General		
Frequency bands		4, 7, 8, 15, 23, 26GHz
Modulation and coding options	Fixed and Adaptive	QPSK, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096 QAM
Channel sizes supported		10, 20, 30, 40, 60, 80 MHz
Capacity range	Airlink Capacity	Up to 920 Mbit/s (single channel WTM 4100) or 1.8 Gbit/s (dual channel operation WTM 4100 with A2C+, or WTM 4200)
	Ethernet / IP Throughput (with IFG/PA Suppression)	Up to 2.5 Gbit/s, single or dual channel, with Multi-Layer Header Compression (actual throughput dependent upon traffic/frame size mix)
Configuration options		1+0 with optional Space Diversity 2+0 Co-Channel Operation with/without XPIC 2+0 Radio Channel Aggregation with Layer 1 Link Aggregation (L1LA) 1+1 or 2+2 Hardware Protection with optional SD 4+0 LAG with/without LOS MIMO (up to 5 Gbit/s)

Transmitter	
High power	Up to +30.5 dBm
Frequency stability	± 5 ppm
Manual transmitter power control range	25 dB
Automatic Transmitter Power Control (ATPC)	Configurable over the full manual attenuation range
Transmitter mute	> 50 dB

Receiver		
Frequency stability		± 5 ppm
Receiver overload	BER=$1x10^{-6}$	20 dBm
Max receiver input level	BER=$1x10^{-3}$	0 dBm
Residual (Background) Bit Error Rate		$1x10^{-13}$

User Interfaces	
Traffic	2x 10/100/1000Base-T (RJ-45) fixed electrical ports (one port supports PoE) 2x optional SFP+ ports – 1, 2.5 or 10 Gbit/s SFP+ (optical/electrical)
DC power supply input	±24/±48 Vdc (SELV) wide-mouth
Console maintenance ports	USB
Receive signal indicator	Dual voltmeter pins

Synchronization	
Synchronous Ethernet (SyncE)	ITU-T G.8262
ESMC/SSM	ITU-T G.8264
Precision Time Protocol	IEEE 1588v2 – TC/BC ITU-T G.8271.1

Carrier Ethernet / L2 Services	
Non-blocking switch	
QoS	8 COS Scheduling, Policing, Storm control, Shaping
QOS mapping	PCP (802.1p), DSCP, H-QoS
VLANs	IEEE 802.1Q IEEE 802.1ad (Q-in-Q)
Spanning tree protocols	STP, RSTP, MSTP
L2 link aggregation	802.1AX
Ethernet ring protection	ITU-T G.8032v2
Ethernet OAM	IEEE 802.1ag, ITU-T Y.1731[1]
Congestion avoidance	RED & WRED
Jumbo frames	to 10 KB

IP / MPLS Services	
IP addressing	IPv4 & IPv6
Routing type	Unicast routing with IP Static routing Equal Cost Multipath IP Routing and MPLS load sharing Segment Routing with OSPF[1] IPv6 for management (OSPFv3[1])
Gateway protocols	IS-IS, OSPF (interior) BGP with Route Reflector, iBGP, eBGP (exterior)
Signaling	Label Distribution Protocol (LDP) and T-LDP LSP protection with BFD Micro-BFD over LAG
Services supported	Layer 2 VPN services over LDP signaled tunnels (VPLS and VPWS) Layer 3 VPN services are supported using BGP
Diagnostics	TWAMP, VRF & LSP Ping and Traceroute
Traffic engineering	Supports RSVP-TE extensions [1]

Element and Network Management	
Local configuration via USB	Configuration save & load, backup to memory stick Wireless dongle to support Bluetooth/Wi-Fi (optional) Aviat OS software upgrade
Event and alarm capture	Time stamp and logging
Statistics	RMON 1 Ethernet and radio performance statistics
Network management	ProVision, ProVision Plus NETCONF/Yang SNMPv2c MIB interface support, SNMPv3 option
IPv4/6 addressing with an in-band management VLAN	SSH access
Simple Network Time Protocol (SNTP V4)	Embedded real time clock
Authentication, Authorisation and Accounting (AAA)	TACACS+, RADIUS (with RadSec) [1]

Mechanical and Environmental		
Operating temperature	Guaranteed Extended	-27° to +131°F (-33° to +55°C) -49° to +149°F (-45° to +65°C)
Humidity	Guaranteed	0 to 100%, non-condensing
Altitude	Guaranteed	14,763 ft (4500 m)
Input voltage		±24/±48 Vdc, wide-mouth (±20 to ±57Vdc)
Power over Ethernet		IEEE 802.3at, 802.3bt
Power consumption	WTM 4100 WTM 4200 WTM 4500	50 W nominal, 65 W maximum 75-90 W nominal, 108 W maximum 58-90 W nominal, 108 W maximum
Size (h-w-d), including built-in coupler/OMT		11.6 in x 10.6 in x 3.75 in (295 mm x 270 mm x 95 mm)
Weight, including coupler/OMT	WTM 4100 WTM 4200 WTM 4500	12 lb (5.5 kg) 17.5 lb (8 kg) 20 lb (9 kg)

Standards Compliance			
Security	FIPS 140-3 Validated, Certificate #5107 FIPS 197, Certificate #A980 AES256 Payload Encryption DISA DoDIN APL Tracking Number (TN) 2503001		
EMC	FCC CFR47 Part 15 ICES-003		
Operation	EN 300 019 Class 4.1		
Safety	UL 62368-1		
RF performance	FCC CFR47 Part 101 NTIA Red Book		
Water ingress	IEC 60529, IP66		
Lightning protection (internal)	IEC 61000-4-5		
Government Certifications		NTIA	DoD
	4 GHz	SPS-27145/1	J/F 12/12222
	7 GHz	SPS-27305/1	J/F 12/12223
	8 GHz	SPS-27298/1	J/F 12/12224
	15 GHz	SPS-27299/1	J/F 12/12225
	26 GHz	SPS-27301/1	J/F 12/12226

Notes:
[1] To be confirmed in a later release

Disclaimer

This material is for informational purposes only and does not constitute a legal obligation to deliver any product, feature or functionality and should not be relied upon in making purchasing decisions. All specifications are nominal across the rated operating temperature range, referenced to the antenna port unless otherwise stated, and are subject to change without notice. The development, release and timing of any features or functionality described for our products is at Aviat Networks' sole discretion.
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